Cancellation of all three-ρ terms $\propto \log(\vee/\Lambda)$

Result

- **All 152 terms cancel exactly.** After symmetrisation of the charges they form 40 sets of identical colour structures, and every set adds up to zero.
- **The cancellation is term against term.** The terms of a set have the same charges, rotated by the same chain of Wilson lines; they differ only in the order of the indices of f^{abc}, i.e. in a sign. No identity is used.
- **Each class and each number of Wilson lines cancels separately.** The four classes $\alpha, \alpha', \beta, \beta'$ do not mix, and each set contains terms with a single number of Wilson lines. So the two-, three- and four-Wilson-line terms cancel among themselves.
- **Numerical check.** With random SU(3) Wilson lines and charges, every kernel monomial has a residual below 3×10^{-15} (Sec. 2).
- **Same as the corrected list.** The list equals the corrected list of `ThreeRho_final.pdf`. It uses your convention $U^{bc} = U^{cb}$; for a general U the cancellation holds when every chain is read as $U(p_1)\cdots U(p_n)\rho$ (oriented from f to the charge).

How to read the equations.

- Every term of the list carries f^{abc} and the common prefactor, which are written in front. The kernel is common to all terms of a class. The overall minus sign of the four-ρ block d^3N_a is absorbed into the signs of its terms.
- Each term is written as in the list, one per line, with its source and its position in that block in grey at the right. $4\rho_a$, $4\rho_b$: from four ρ's (d^3N_a, d^3N_b); I.II.1: Group I Row II part 1; and so on.
- Terms of the same colour cancel each other. The last lines of each equation give, for every colour, the sum of the coefficients times the common colour structure; it is zero. There $(\cdots)$ stands for the prefactor and the kernel of the first line.
- The charges are compared in their fully symmetrised form, so $\frac{1}{2}\rho\{\rho,\rho\}$ counts as $\rho\rho\rho$ and $\{\rho,\rho\}\rho$ as $2\rho\rho\rho$. A term with its f indices in another order picks up the sign of the reordering.
- In the sums, $\big[U(w)U(w')\rho(x')\big]^c = U^{cd}(w)U^{de}(w')\rho^e(x')$.

class	Wilson lines	terms	sets (each sums to 0)
α	2	4	2
α	3	18	4
α	4	6	2
α'	2	4	2
α'	3	18	4
α'	4	6	2
β	2	6	2
β	3	36	8
β	4	6	2
β'	2	6	2
β'	3	36	8
β'	4	6	2
total		152	40

1 Cancellation of all three ρ's

1.1 α terms

1.1.1 Two Wilson lines

4 terms in 2 colours.

$$\begin{aligned}
\frac{d^3N_{2\,U}^{(\alpha)}}{d^3k} &= \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{\Lambda}\right)\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\\
&\quad\times\Big[-\rho^d(x')\rho^a(y)\rho^c(x)\,U^{de}(w')U^{eb}(w) && 4\rho_a\,(1)\\
&\quad\ -\tfrac{1}{2}\left\{\rho^e(x'),\rho^b(y)\right\}\rho^c(x)\,U^{de}(w')\,U^{ad}(w) && \text{II.I}\,(1)\\
&\quad\ +\rho^e(x')\rho^a(y)\rho^c(x)\,U^{db}(w)U^{ed}(x') && 4\rho_a\,(2)\\
&\quad\ +\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\Big] && \text{III.II}\,(3)\\
&= (\cdots)\Big[\left(-1+1\right)\rho^a(x)\,\rho^b(y)\left[U(w)U(w')\rho(x')\right]^c\\
&\quad\ \left(1-1\right)\rho^a(x)\,\rho^b(y)\left[U(w)U(x')\rho(x')\right]^c\Big]=0. && (1)
\end{aligned}$$

1.1.2 Three Wilson lines

18 terms in 4 colours.

$$\begin{aligned}
\frac{d^3N_{3\,U}^{(\alpha)}}{d^3k} &= \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{\Lambda}\right)\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\\
&\quad\times\Big[-\rho^d(x')\rho^h(y)\rho^e(x)U^{db}(w')U^{ec}(x)U^{ah}(y) && 4\rho_b\,(1)\\
&\quad\ +\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y) && \text{I.II.1}\,(3)\\
&\quad\ -\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x) && \text{I.II.1}\,(4)\\
&\quad\ +\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y) && \text{I.III}\,(4)\\
&\quad\ +\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(w')\,U^{be}(y)\,U^{ch}(x) && \text{I.IV}\,(4)\\
&\quad\ -\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x) && \text{III.I}\,(4)\\
&\quad\ +\rho^e(x')\rho^h(y)\rho^d(x)U^{dc}(x)U^{be}(x')U^{ah}(y) && 4\rho_b\,(4)\\
&\quad\ -\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y) && \text{I.II.1}\,(1)\\
&\quad\ +\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x) && \text{I.II.1}\,(2)\\
&\quad\ -\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y) && \text{I.III}\,(2)\\
&\quad\ -\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x) && \text{I.IV}\,(2)\\
&\quad\ -\tfrac{1}{2}\left\{\rho^d(x'),\rho^e(y)\right\}\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x) && \text{II.I}\,(4)\\
&\quad\ +\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x) && \text{III.I}\,(2)\\
&\quad\ +\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{bd}(y)\,U^{ae}(x')\,U^{ch}(x) && \text{III.II}\,(2)\\
&\quad\ +\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z) && \text{I.III}\,(1)\\
&\quad\ -\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x) && \text{III.II}\,(4)\\
&\quad\ -\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z) && \text{I.III}\,(3)\\
&\quad\ +\tfrac{1}{2}\left\{\rho^d(x'),\rho^e(y)\right\}\rho^h(x)\,U^{be}(z)\,U^{ad}(w')\,U^{ch}(x)\Big] && \text{II.I}\,(2)\\
&= (\cdots)\Big[\left(-1+1+1-1-1+1\right)\left[U(w')\rho(x')\right]^a\left[U(x)\rho(x)\right]^b\left[U(y)\rho(y)\right]^c\\
&\quad\ \left(-1+1+1-1-1-1+1+1\right)\left[U(x)\rho(x)\right]^a\left[U(x')\rho(x')\right]^b\left[U(y)\rho(y)\right]^c\\
&\quad\ \left(1-1\right)\left[U(z)\rho(y)\right]^a\left[U(x)\rho(x)\right]^b\left[U(x')\rho(x')\right]^c\Big]
\end{aligned}$$

$$\left.\Big(-1+1\Big)\left[U(w')\rho(x')\right]^a\left[U(z)\rho(y)\right]^b\left[U(x)\rho(x)\right]^c\right] = 0. \tag{2}$$

1.1.3 Four Wilson lines

6 terms in 2 colours.

$$\begin{aligned}
\frac{d^3N_{4\,U}^{(\alpha)}}{d^3k} &= \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{\Lambda}\right)\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\\
&\times\Bigg[+\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y) && \text{I.IV (1)}\\
&\quad+\tfrac{1}{2}\left\{\rho^h(x'),\rho^l(y)\right\}\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ae}(w) && \text{II.I (3)}\\
&\quad-\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w) && \text{III.I (1)}\\
&\quad-\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ah}(w) && \text{III.II (1)}\\
&\quad-\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y) && \text{I.IV (3)}\\
&\quad+\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)\Bigg] && \text{III.I (3)}\\
&= (\cdots)\Bigg[\Big(1+1-1-1\Big)\,\rho^a(x)\left[U(w)U(x')\rho(x')\right]^b\left[U(z)U(y)\rho(y)\right]^c\\
&\quad\Big(-1+1\Big)\,\rho^a(x)\left[U(w)U(w')\rho(x')\right]^b\left[U(z)U(y)\rho(y)\right]^c\Bigg] = 0.
\end{aligned} \tag{3}$$

1.2 α' terms

1.2.1 Two Wilson lines

4 terms in 2 colours.

$$\begin{aligned}
\frac{d^3N_{2\,U}^{(\alpha')}}{d^3k} &= \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{\Lambda}\right)\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Bigg[-\rho^c(x')\rho^a(y)\rho^e(x)\,U^{db}(w')U^{ed}(x) && 4\rho_a\ (3)\\
&\quad-\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{de}(x)\,U^{ad}(w') && \text{III.II (11)}\\
&\quad+\rho^c(x')\rho^a(y)\rho^e(x)U^{bd}(w')U^{ed}(w) && 4\rho_a\ (4)\\
&\quad+\tfrac{1}{2}\,\rho^c(x')\left\{\rho^e(x),\rho^b(y)\right\}U^{ad}(w')\,U^{de}(w)\Bigg] && \text{II.I (9)}\\
&= (\cdots)\Bigg[\Big(-1+1\Big)\,\rho^a(x')\,\rho^b(y)\left[U(w')U(x)\rho(x)\right]^c\\
&\quad\Big(1-1\Big)\,\rho^a(x')\,\rho^b(y)\left[U(w')U(w)\rho(x)\right]^c\Bigg] = 0.
\end{aligned} \tag{4}$$

1.2.2 Three Wilson lines

18 terms in 4 colours.

$$\begin{aligned}
\frac{d^3N_{3\,U}^{(\alpha')}}{d^3k} &= \frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{\Lambda}\right)\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
&\times\Bigg[+\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{eb}(w)U^{ah}(y) && 4\rho_b\ (2)\\
&\quad-\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y) && \text{I.II.2 (3)}\\
&\quad+\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x') && \text{I.II.2 (4)}\\
&\quad-\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y) && \text{I.III (8)}
\end{aligned}$$

$$
\begin{aligned}
&-\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{da}(w)\,U^{ch}(x')\,U^{be}(y) && \text{I.IV (8)}\\
&+\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x') && \text{III.I (12)}\\
&-\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{be}(x)U^{ah}(y) && 4\rho_b\ (3)\\
&+\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y) && \text{I.II.2 (1)}\\
&-\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x') && \text{I.II.2 (2)}\\
&+\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y) && \text{I.III (6)}\\
&+\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{da}(x)\,U^{ch}(x')\,U^{be}(y) && \text{I.IV (6)}\\
&+\tfrac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y) && \text{II.I (12)}\\
&-\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x') && \text{III.I (10)}\\
&-\rho^h(x')\,\rho^e(x)\,\rho^d(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x') && \text{III.II (10)}\\
&-\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z) && \text{I.III (5)}\\
&+\rho^h(x')\,\rho^d(x)\,\rho^e(y)\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x') && \text{III.II (12)}\\
&+\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z) && \text{I.III (7)}\\
&-\tfrac{1}{2}\,\rho^e(x')\big\{\rho^h(x),\rho^d(y)\big\}\,U^{bd}(z)\,U^{ce}(x')\,U^{ah}(w)\Big] && \text{II.I (11)}
\end{aligned}
$$

$$
\begin{aligned}
=(\cdots)\Big[&(1-1-1+1+1-1)\,\big[U(w)\rho(x)\big]^a\,\big[U(x')\rho(x')\big]^b\,\big[U(y)\rho(y)\big]^c\\
&(-1+1+1-1-1-1+1+1)\,\big[U(x)\rho(x)\big]^a\,\big[U(x')\rho(x')\big]^b\,\big[U(y)\rho(y)\big]^c\\
&(1-1)\,\big[U(z)\rho(y)\big]^a\,\big[U(x)\rho(x)\big]^b\,\big[U(x')\rho(x')\big]^c\\
&(1-1)\,\big[U(w)\rho(x)\big]^a\,\big[U(z)\rho(y)\big]^b\,\big[U(x')\rho(x')\big]^c\Big]=0.
\end{aligned}
\tag{5}
$$

1.2.3 Four Wilson lines

6 terms in 2 colours.

$$
\begin{aligned}
\frac{d^3N_{4\,U}^{(\alpha')}}{d^3k}=\frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{\Lambda}\Big)\int_{x,y,z,x',w,w'}e^{-ik(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\
\times\Big[-\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y) && \text{I.IV (5)}\\
-\tfrac{1}{2}\,\rho^c(x')\big\{\rho^h(x),\rho^l(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y) && \text{II.I (10)}\\
+\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w') && \text{III.I (9)}\\
+\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ah}(w') && \text{III.II (9)}\\
+\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y) && \text{I.IV (7)}\\
-\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w)\,U^{ha}(w')\Big] && \text{III.I (11)}
\end{aligned}
$$

$$
\begin{aligned}
=(\cdots)\Big[&(-1-1+1+1)\,\rho^a(x')\,\big[U(w')U(x)\rho(x)\big]^b\,\big[U(z)U(y)\rho(y)\big]^c\\
&(1-1)\,\rho^a(x')\,\big[U(w')U(w)\rho(x)\big]^b\,\big[U(z)U(y)\rho(y)\big]^c\Big]=0.
\end{aligned}
\tag{6}
$$

1.3 β terms

1.3.1 Two Wilson lines

6 terms in 2 colours.

$$
\begin{aligned}
\frac{d^3N_{2\,U}^{(\beta)}}{d^3k}=\frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{\Lambda}\Big)\int_{x,y,z,x',w,w'}e^{-ik(w'-w)}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\\
\times\Big[-\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(x')\,U^{ad}(w) && \text{I.II.1 (5)}
\end{aligned}
$$

$$
\begin{aligned}
&+\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w) && \text{I.II.1 (6)}\\
&-2\,\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{da}(w) && \text{III.II (7)}\\
&+\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(w')\,U^{ad}(w) && \text{I.II.1 (9)}\\
&-\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w) && \text{I.II.1 (10)}\\
&+\big\{\rho^e(x'),\rho^b(y)\big\}\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\Big] && \text{II.I (5)}\\
=(\cdots)&\Big[(-1-1+2)\,\rho^a(x)\,\rho^b(y)\,\big[U(w)U(x')\rho(x')\big]^c\\
&(1+1-2)\,\rho^a(x)\,\rho^b(y)\,\big[U(w)U(w')\rho(x')\big]^c\Big]=0. && (7)
\end{aligned}
$$

1.3.2 Three Wilson lines

36 terms in 8 colours.

$$
\begin{aligned}
\frac{d^3N_{3\,U}^{(\beta)}}{d^3k}=\frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{\Lambda}\Big)\int_{x,y,z,x',w,w'}e^{-ik(w'-w)}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\\
\times\Big[+\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y) && \text{I.II.1 (7)}\\
-\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x) && \text{I.II.1 (8)}\\
+2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x) && \text{I.III (10)}\\
+2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{hc}(x) && \text{I.IV (11)}\\
+\tfrac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y) && \text{I.V.1 (3)}\\
-\tfrac{1}{2}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x) && \text{I.V.2 (2)}\\
+\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x) && \text{II.I (8)}\\
-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x) && \text{III.I (6)}\\
-2\,\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{db}(y)\,U^{ae}(x')\,U^{ch}(x) && \text{III.II (6)}\\
+\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y) && \text{III.III (4)}\\
-2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x') && \text{III.V (3)}\\
-2\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{bh}(y)\,U^{ad}(x)\,U^{ce}(x') && \text{III.VI (3)}\\
-\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y) && \text{I.II.1 (11)}\\
+\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x) && \text{I.II.1 (12)}\\
-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x) && \text{I.III (12)}\\
-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(w')\,U^{be}(y)\,U^{hc}(x) && \text{I.IV (12)}\\
-\tfrac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y) && \text{I.V.1 (4)}\\
+\tfrac{1}{2}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x) && \text{I.V.2 (4)}\\
+2\,\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x) && \text{III.I (8)}\\
+2\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(w')\,U^{bh}(y) && \text{III.V (4)}\\
-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(w) && \text{I.III (9)}\\
-\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y) && \text{III.III (2)}\\
+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x') && \text{III.V (1)}\\
+2\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y) && \text{III.VI (1)}\\
+2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(w) && \text{I.III (11)}\\
-2\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(w')\,U^{bh}(y) && \text{III.V (2)}\\
-\tfrac{1}{2}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w) && \text{I.V.1 (1)}\\
+\tfrac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{be}(w)\,U^{ch}(z) && \text{I.V.2 (1)}\\
-2\,\rho^e(y)\,\rho^h(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(x') && \text{III.VI (2)}
\end{aligned}
$$

$$
\begin{aligned}
&+\tfrac{1}{2}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w) && \text{I.V.1 (2)}\\
&-\tfrac{1}{2}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{be}(w)\,U^{ch}(z) && \text{I.V.2 (3)}\\
&+\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w') && \text{III.III (1)}\\
&-\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{ad}(w')\,U^{eb}(z)\,U^{ch}(x) && \text{II.I (6)}\\
&-\big\{\rho^h(x'),\rho^d(y)\big\}\rho^e(x)\,U^{db}(z)\,U^{ae}(x)\,U^{ch}(w') && \text{III.III (3)}\\
&+2\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x) && \text{III.II (8)}\\
&+2\,\rho^d(y)\,\rho^h(x')\,\rho^e(x)\,U^{ae}(x)\,U^{ch}(x')\,U^{db}(z)\Big] && \text{III.VI (4)}
\end{aligned}
$$

$$
\begin{aligned}
=(\cdots)\Big[&(-1-1-2+2+1+1+2-2-2-2+2+2)\,\big[U(x)\rho(x)\big]^a\,\big[U(x')\rho(x')\big]^b\,\big[U(y)\rho(y)\big]^c\\
&(-1-1-2+2+1+1-2+2)\,\big[U(w')\rho(x')\big]^a\,\big[U(x)\rho(x)\big]^b\,\big[U(y)\rho(y)\big]^c\\
&(2+2-2-2)\,\big[U(w)\rho(x)\big]^a\,\big[U(x')\rho(x')\big]^b\,\big[U(y)\rho(y)\big]^c\\
&(-2+2)\,\big[U(w)\rho(x)\big]^a\,\big[U(w')\rho(x')\big]^b\,\big[U(y)\rho(y)\big]^c\\
&(1+1-2)\,\big[U(w)\rho(x)\big]^a\,\big[U(z)\rho(y)\big]^b\,\big[U(x')\rho(x')\big]^c\\
&(1+1-2)\,\big[U(w)\rho(x)\big]^a\,\big[U(w')\rho(x')\big]^b\,\big[U(z)\rho(y)\big]^c\\
&(-2+2)\,\big[U(w')\rho(x')\big]^a\,\big[U(z)\rho(y)\big]^b\,\big[U(x)\rho(x)\big]^c\\
&(2-2)\,\big[U(z)\rho(y)\big]^a\,\big[U(x)\rho(x)\big]^b\,\big[U(x')\rho(x')\big]^c\Big]=0.
\end{aligned}
\tag{8}
$$

1.3.3 Four Wilson lines

6 terms in 2 colours.

$$
\begin{aligned}
\frac{d^3N_{4\,U}^{(\beta)}}{d^3k}=\frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{\Lambda}\Big)\int_{x,y,z,x',w,w'}e^{-ik(w'-w)}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\\
\times\Big[-2\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y) && \text{I.IV (9)}\\
-\big\{\rho^h(x'),\rho^l(y)\big\}\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ea}(w) && \text{II.I (7)}\\
+2\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w) && \text{III.I (5)}\\
+2\,\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ha}(w) && \text{III.II (5)}\\
+2\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y) && \text{I.IV (10)}\\
-2\,\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)\Big] && \text{III.I (7)}
\end{aligned}
$$

$$
\begin{aligned}
=(\cdots)\Big[&(-2-2+2+2)\,\rho^a(x)\,\big[U(w)U(x')\rho(x')\big]^b\,\big[U(z)U(y)\rho(y)\big]^c\\
&(2-2)\,\rho^a(x)\,\big[U(w)U(w')\rho(x')\big]^b\,\big[U(z)U(y)\rho(y)\big]^c\Big]=0.
\end{aligned}
\tag{9}
$$

1.4 β' terms

1.4.1 Two Wilson lines

6 terms in 2 colours.

$$
\begin{aligned}
\frac{d^3N_{2\,U}^{(\beta')}}{d^3k}=\frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{\Lambda}\Big)\int_{x,y,z,x',w,w'}e^{-ik(w'-w)}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\\
\times\Big[+\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w') && \text{I.II.2 (5)}\\
-\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w') && \text{I.II.2 (6)}
\end{aligned}
$$

$$
\begin{aligned}
&+2\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{de}(x)\,U^{ad}(w') && \text{III.II (15)}\\
&-\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w') && \text{I.II.2 (9)}\\
&+\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w') && \text{I.II.2 (10)}\\
&-\rho^c(x')\big\{\rho^e(x),\rho^b(y)\big\}\,U^{ad}(w')\,U^{de}(w)\Big] && \text{II.I (13)}\\
=(\cdots)\Big[&(1+1-2)\,\rho^a(x')\,\rho^b(y)\,\big[U(w')U(x)\rho(x)\big]^c\\
&\big(-1-1+2\big)\,\rho^a(x')\,\rho^b(y)\,\big[U(w')U(w)\rho(x)\big]^c\Big]=0. && (10)
\end{aligned}
$$

1.4.2 Three Wilson lines

36 terms in 8 colours.

$$
\begin{aligned}
\frac{d^3N_{3\,U}^{(\beta')}}{d^3k}=\frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{\Lambda}\Big)\int_{x,y,z,x',w,w'}e^{-ik(w'-w)}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\\
\times\Big[&-\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y) && \text{I.II.2 (7)}\\
&+\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x') && \text{I.II.2 (8)}\\
&-2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x') && \text{I.III (14)}\\
&-2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x') && \text{I.IV (16)}\\
&-\tfrac{1}{2}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{bh}(y)\,U^{ce}(x') && \text{I.V.1 (8)}\\
&+\tfrac{1}{2}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x') && \text{I.V.2 (6)}\\
&-\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y) && \text{II.I (16)}\\
&+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x') && \text{III.I (14)}\\
&+2\,\rho^h(x')\,\rho^e(x)\,\rho^d(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x') && \text{III.II (14)}\\
&-\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y) && \text{III.III (8)}\\
&+2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{be}(y)\,U^{ad}(x')\,U^{ch}(x) && \text{III.V (7)}\\
&+2\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{bh}(y)\,U^{ad}(x')\,U^{ce}(x) && \text{III.VI (7)}\\
&+\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y) && \text{I.II.2 (11)}\\
&-\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x') && \text{I.II.2 (12)}\\
&+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(x') && \text{I.III (16)}\\
&+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(w)\,U^{ch}(x') && \text{I.IV (15)}\\
&+\tfrac{1}{2}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(y)\,U^{ce}(x') && \text{I.V.1 (7)}\\
&-\tfrac{1}{2}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(x')\,U^{ce}(y) && \text{I.V.2 (8)}\\
&-2\,\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x') && \text{III.I (16)}\\
&-2\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(x')\,U^{ce}(w)\,U^{bh}(y) && \text{III.V (8)}\\
&+2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(w') && \text{I.III (13)}\\
&+\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y) && \text{III.III (6)}\\
&-2\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x) && \text{III.V (5)}\\
&-2\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{bh}(y)\,U^{ce}(x) && \text{III.VI (5)}\\
&-2\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(w') && \text{I.III (15)}\\
&+2\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(w')\,U^{ce}(w)\,U^{bh}(y) && \text{III.V (6)}\\
&+\tfrac{1}{2}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(w') && \text{I.V.1 (5)}\\
&-\tfrac{1}{2}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{be}(w')\,U^{ch}(z) && \text{I.V.2 (5)}\\
&+2\,\rho^d(x')\,\rho^h(x)\,\rho^e(y)\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(x) && \text{III.VI (6)}\\
&-\tfrac{1}{2}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w') && \text{I.V.1 (6)}
\end{aligned}
$$

$$
\begin{aligned}
&+\tfrac{1}{2}\,\{\rho^e(x'),\rho^h(y)\}\rho^d(x)\,U^{ad}(w)\,U^{ch}(z)\,U^{be}(w') && \text{I.V.2 (7)}\\
&-\rho^d(x')\{\rho^h(x),\rho^e(y)\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w) && \text{III.III (5)}\\
&+\rho^e(x')\{\rho^h(x),\rho^d(y)\}\,U^{ce}(x')\,U^{db}(z)\,U^{ah}(w) && \text{II.I (15)}\\
&+\rho^e(x')\{\rho^h(x),\rho^d(y)\}\,U^{bd}(z)\,U^{ae}(x')\,U^{ch}(w) && \text{III.III (7)}\\
&-2\,\rho^h(x')\,\rho^d(x)\,\rho^e(y)\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x') && \text{III.II (16)}\\
&-2\,\rho^e(x')\,\rho^h(x)\,\rho^d(y)\,U^{ae}(x')\,U^{ch}(x)\,U^{bd}(z)\Big] && \text{III.VI (8)}
\end{aligned}
$$

$$
\begin{aligned}
=(\cdots)\Big[&(-1-1-2+2+1+1+2-2-2-2+2+2)\,[U(x)\rho(x)]^a\,[U(x')\rho(x')]^b\,[U(y)\rho(y)]^c\\
&(1+1+2-2-1-1+2-2)\,[U(w)\rho(x)]^a\,[U(x')\rho(x')]^b\,[U(y)\rho(y)]^c\\
&(-2-2+2+2)\,[U(w')\rho(x')]^a\,[U(x)\rho(x)]^b\,[U(y)\rho(y)]^c\\
&(-2+2)\,[U(w)\rho(x)]^a\,[U(w')\rho(x')]^b\,[U(y)\rho(y)]^c\\
&(-1-1+2)\,[U(w')\rho(x')]^a\,[U(z)\rho(y)]^b\,[U(x)\rho(x)]^c\\
&(1+1-2)\,[U(w)\rho(x)]^a\,[U(w')\rho(x')]^b\,[U(z)\rho(y)]^c\\
&(2-2)\,[U(w)\rho(x)]^a\,[U(z)\rho(y)]^b\,[U(x')\rho(x')]^c\\
&(2-2)\,[U(z)\rho(y)]^a\,[U(x)\rho(x)]^b\,[U(x')\rho(x')]^c\Big]=0.
\end{aligned}
\tag{11}
$$

1.4.3 Four Wilson lines

6 terms in 2 colours.

$$
\begin{aligned}
\frac{d^3N_{4\,U}^{(\beta')}}{d^3k}=\frac{i\,f^{abc}}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{\Lambda}\Big)\int_{x,y,z,x',w,w'}e^{-ik(w'-w)}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\\
\times\Big[+2\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y) && \text{I.IV (13)}\\
+\rho^c(x')\{\rho^h(x),\rho^l(y)\}\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y) && \text{II.I (14)}\\
-2\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w') && \text{III.I (13)}\\
-2\,\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ha}(w') && \text{III.II (13)}\\
-2\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y) && \text{I.IV (14)}\\
+2\,\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{de}(y)\,U^{hl}(w)\,U^{db}(z)\,U^{ha}(w')\Big] && \text{III.I (15)}
\end{aligned}
$$

$$
\begin{aligned}
=(\cdots)\Big[&(2+2-2-2)\,\rho^a(x')\,[U(w')U(x)\rho(x)]^b\,[U(z)U(y)\rho(y)]^c\\
&(-2+2)\,\rho^a(x')\,[U(w')U(w)\rho(x)]^b\,[U(z)U(y)\rho(y)]^c\Big]=0.
\end{aligned}
\tag{12}
$$

2 Numerical check

Each term is reduced to its symmetrised charge part. The three charges are placed on three points in all 3! ways, the terms are collected by kernel monomial, and the colour factors are evaluated with random SU(3) adjoint Wilson lines and random charges, in three configurations (`check_cancel.py`).

Wilson lines	set	kernel monomials	largest residual (largest single term)
$U = U^T$ (your convention)	all	24	$2.4e-15$ ($2.2e+00$)
$U = U^T$ (your convention)	α	6	$1.9e-15$ ($1.0e+00$)
$U = U^T$ (your convention)	α'	6	$1.9e-15$ ($1.1e+00$)
$U = U^T$ (your convention)	β	6	$2.4e-15$ ($2.1e+00$)
$U = U^T$ (your convention)	β'	6	$2.2e-15$ ($2.2e+00$)
general U, as written	all	24	$6.0e+00$ ($2.8e+00$)
general U, as written	α	6	$4.6e+00$ ($1.4e+00$)
general U, as written	α'	6	$4.5e+00$ ($1.1e+00$)
general U, as written	β	6	$6.0e+00$ ($2.3e+00$)
general U, as written	β'	6	$3.6e+00$ ($2.8e+00$)

For a general U, with every chain read as $U(p_1)\cdots U(p_n)\rho$ (oriented from f to the charge), the sum over each class vanishes to 10^{-14} (`general_u.py`).